

Research Notes on Instruction tuning

Instruction tuning

>> Section 1

Architecture LLMs are generally based on the transformer architecture, which leverages an attention mechanism that enables the model to process relationships between all elements in a sequence simultaneously, regardless of their distance from each other.

>> Section 2

Legal and commercial responses to memorization and training-data practices have accelerated, producing a mix of rulings, ongoing suits, and large settlements that turn on factual details such as how data were acquired and retained and whether use for model training is sufficiently "transformative" to qualify as fair use. In 2025, Anthropic reached a preliminary agreement to settle a class action by authors for about \$1.5 billion after a judge found the company had stored millions of pirated books in a library, despite the judge describing aspects of training as transformative. Meta obtained a favorable judgment in mid-2025 in a suit by thirteen authors after the court found the plaintiffs had not developed a record sufficient to show infringement in that limited case. OpenAI continues to face multiple suits by authors and news organizations with mixed procedural outcomes and contested evidentiary issues. Memorization was an emergent behavior in early, completion language models in which long strings of text are occasionally output verbatim from training data, contrary to the typical behavior of traditional artificial neural networks. Evaluations of controlled LLM output measure the amount memorized from training data (focused on GPT-2-series models) as variously over 1% for exact duplicates or up to about 7%. A 2023 study showed that when ChatGPT 3.5 turbo was prompted to repeat the same word indefinitely, after a few hundreds of repetitions, it would start outputting excerpts from its training data.

>> Section 3

In order to find out which tokens are relevant to each other within the scope of the context window, the attention mechanism calculates "soft" weights for each token, more precisely for its embedding, by using multiple attention heads, each with its own "relevance" for calculating its own soft weights. For example, the small (i.e. 117M parameter sized) GPT-2 model has had twelve attention heads and a context window of only 1k tokens. In its medium version it has 345M parameters and contains 24 layers, each with 12 attention heads. For the training with gradient descent a batch size of 512 was utilized. Autoregressive models, such as GPTs, are trained to guess how a sequence continues; for example, whether the word sequence "I like to eat" is more likely to be followed by the word "bread" or the word "rocks". Masked models, such as BERT, are

trained to guess parts that are missing from a sequence, such as whether the missing word in "I like to ___ roses" is more likely to be the word "smell" or the word "eat". The model's predictions are based on the properties of sequences within its training dataset.

>> Section 4

Sleeper agents Researchers from Anthropic found that it was possible to create "sleeper agents", models with hidden functionalities that remain dormant until triggered by a specific event or condition. Upon activation, the LLM deviates from its expected behavior to make insecure actions. For example, an LLM could produce safe code except on a specific date, or if the prompt contains a specific tag. These functionalities were found to be difficult to detect or remove via safety training.

>> Section 5

Benchmarks are used to evaluate LLM performance on specific tasks. Tests evaluate capabilities such as general knowledge, bias, commonsense reasoning, question answering, and mathematical problem-solving. Composite benchmarks examine multiple capabilities. Results are often sensitive to the prompting method. LLM bias may be assessed through benchmarks such as CrowS-Pairs (Crowdsourced Stereotype Pairs), Stereo Set, and Parity Benchmark. Fact-checking and misinformation detection benchmarks are available. A 2023 study compared the fact-checking accuracy of LLMs including ChatGPT 3.5 and 4.0, Bard, and Bing AI against independent fact-checkers such as PolitiFact and Snopes. The results demonstrated moderate proficiency, with GPT-4 achieving the highest accuracy at 71%, lagging behind human fact-checkers. In addition to standard NLP benchmarks, LLMs have been evaluated as substitutes for human annotators. Several studies find that models such as GPT-3.5 and GPT-4 can outperform crowd workers or student coders on a range of text-annotation tasks, including moderation and classification of political content in English and Spanish news.

>> Section 6

NLP researchers were evenly split when asked, in a 2022 survey, whether (untuned) LLMs "could (ever) understand natural language in some nontrivial sense". Proponents of "LLM understanding" believe that some LLM abilities, such as mathematical reasoning, imply an ability to "understand" certain concepts. A Microsoft team argued in 2023 that GPT-4 "can solve novel and difficult tasks that span mathematics, coding, vision, medicine, law, psychology and more" and that GPT-4 "could reasonably be viewed as an early (yet still incomplete) version of an artificial general intelligence system": "Can one reasonably say that a

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system that passes exams for software engineering candidates is not really intelligent?" Ilya Sutskever argues that predicting the next word sometimes involves reasoning and deep insights, for example if the LLM has to predict the name of the criminal in an unknown detective novel after processing the entire story leading up to the revelation. Some researchers characterize LLMs as "alien intelligence". For example, Conjecture CEO Connor Leahy considers untuned LLMs to be like inscrutable alien "Shoggoths", and believes that RLHF tuning creates a "smiling facade" obscuring the inner workings of the LLM: "If you don't push it too far, the smiley face stays on. But then you give it [an unexpected] prompt, and suddenly you see this massive underbelly of insanity, of weird thought processes and clearly non-human understanding." In contrast, some skeptics of LLM understanding believe that existing LLMs are "simply remixing and recombining existing writing", a phenomenon known as stochastic parrot, or they point to the deficits existing LLMs continue to have in prediction skills, reasoning skills, agency, and explainability. For example, GPT-4 has natural deficits in planning and in real-time learning. Generative LLMs have been observed to confidently assert claims of fact which do not seem to be justified by their training data, a phenomenon which has been termed "hallucination". Specifically, hallucinations in the context of LLMs correspond to the generation of text or responses that seem syntactically sound, fluent, and natural but are factually incorrect, nonsensical, or unfaithful to the provided source input. Neuroscientist Terrence Sejnowski has argued that "The diverging opinions of experts on the intelligence of LLMs suggests that our old ideas based on natural intelligence are inadequate". Efforts to reduce or compensate for hallucinations have employed automated reasoning, retrieval-augmented generation (RAG), fine-tuning, and other methods. The matter of LLM's exhibiting intelligence or understanding has two main aspects—the first is how to model thought and language in a computer system, and the second is how to enable the computer system to generate human-like language. These aspects of language as a model of cognition have been developed in the field of cognitive linguistics. American linguist George Lakoff presented neural theory of language (NTL) as a computational basis for using language as a model of learning tasks and understanding. The NTL model outlines how specific neural structures of the human brain shape the nature of thought and language and in turn what are the computational properties of such neural systems that can be applied to model thought and language in a computer system. After a framework for modeling language in a computer systems was established, the focus shifted to establishing frameworks for computer systems to generate language with

acceptable grammar. In his 2014 book titled *The Language Myth: Why Language Is Not An Instinct*, British cognitive linguist and digital communication technologist Vyvyan Evans mapped out the role of probabilistic context-free grammar (PCFG) in enabling NLP to model cognitive patterns and generate human-like language.